# Analogue electronics: mixed questions

National 5 · Electronics & control · Answers

## Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

**1 A 12 V supply is connected to a 240 Ω resistor. Calculate the current.**
- I = 12 ÷ 240 = 0.05 A

**2a) Calculate the total resistance.**
- $R_T$ = 50 Ω

**2b) Calculate the current.**
- I = 10 ÷ 50 = 0.2 A

**2c) Calculate the voltage across each resistor.**
- 3 V and 7 V

**3a) Calculate the total resistance.**
- $R_T$ = 75 Ω

**3b) Calculate the total current.**
- $I_T$ = 12 ÷ 75 = 0.16 A

**3c) Calculate the current through the 100 Ω resistor.**
- 12 ÷ 100 = 0.12 A

**4 A current of 0.5 A flows through a resistor on a 9 V supply. Calculate the resistance.**
- R = 9 ÷ 0.5 = 18 Ω

**5a) Calculate the total resistance.**
- $R_{parallel}$ = 20 Ω; $R_T$ = 40 Ω

**5b) Calculate the current.**
- I = 18 ÷ 40 = 0.45 A

**5c) Calculate the voltage across the 20 Ω resistor.**
- V = 0.45 × 20 = 9 V

**6a) Calculate the total resistance.**
- 6.9 kΩ

**6b) Calculate the current.**
- I = 12 ÷ 6900 = 1.74 mA

**6c) Calculate the voltage across $R_2$.**
- V = 0.00174 × 4700 = 8.2 V

**7a) Calculate the total resistance.**
- $1/R_T$ = 0.1 + 0.05 + 0.025 = 0.175; $R_T$ = 5.71 Ω

**7b) Calculate the total current.**
- $I_T$ = 12 ÷ 5.71 = 2.1 A

**7c) Calculate the current through the 20 Ω resistor.**
- 12 ÷ 20 = 0.6 A

**8 A current of 2 A flows through a 15 Ω resistor. Calculate the supply voltage.**
- V = 2 × 15 = 30 V

**9a) Calculate the total resistance.**
- 500 Ω

**9b) Calculate the current.**
- 20 ÷ 500 = 0.04 A

**9c) Calculate the voltage across the 250 Ω resistor.**
- 0.04 × 250 = 10 V

**10a) Calculate the total resistance.**
- (8 × 12) ÷ 20 = 4.8 Ω

**10b) Calculate the total current.**
- 6 ÷ 4.8 = 1.25 A

**11a) Calculate the total resistance.**
- $R_{parallel}$ = 50 Ω; $R_T$ = 100 Ω

**11b) Calculate the current.**
- 10 ÷ 100 = 0.1 A

**11c) Calculate the current through one of the 100 Ω resistors.**
- $V_{50}$ = 5 V, so 5 V across the pair
- I = 5 ÷ 100 = 0.05 A

**12a) Draw a circuit to give 25 Ω.**
- The two 50 Ω resistors in parallel

**12b) Prove your answer with a calculation.**
- (50 × 50) ÷ 100 = 25 Ω

**13 A 24 V supply is connected to a 3 kΩ resistor. Calculate the current in mA.**
- 24 ÷ 3000 = 0.008 A = 8 mA

**14a) Calculate the total resistance.**
- (1000 × 2000) ÷ 3000 = 667 Ω

**14b) Calculate the total current.**
- 9 ÷ 667 = 0.0135 A

**15a) Calculate the total resistance.**
- $R_{parallel}$ = 20 Ω; $R_T$ = 30 Ω

**15b) Calculate the current.**
- 12 ÷ 30 = 0.4 A

**15c) Calculate the voltage across the 10 Ω resistor.**
- 0.4 × 10 = 4 V

**15d) Calculate the current through the 30 Ω resistor.**
- $V_{parallel}$ = 12 − 4 = 8 V
- I = 8 ÷ 30 = 0.27 A

**16a) Calculate the total resistance.**
- 1970 Ω

**16b) Calculate the current.**
- 9 ÷ 1970 = 4.57 mA

**17a) Calculate the total resistance.**
- (20 × 30) ÷ 50 = 12 Ω

**17b) Calculate the total current.**
- 12 ÷ 12 = 1 A

**17c) Calculate the current through the 20 Ω heater.**
- 12 ÷ 20 = 0.6 A

**18 A circuit has a total resistance of 40 Ω: a 10 Ω resistor in series with a parallel section. Calculate the resistance of the parallel section.**
- 40 − 10 = 30 Ω

**19a) Calculate the total resistance.**
- 1250 Ω

**19b) Calculate the total current.**
- 10 ÷ 1250 = 8 mA

**19c) Calculate the current through the 2 kΩ resistor.**
- 10 ÷ 2000 = 5 mA

**20a) Calculate the total resistance.**
- $R_{parallel}$ = 15 Ω; $R_T$ = 30 Ω

**20b) Calculate the current.**
- 6 ÷ 30 = 0.2 A

**20c) Calculate the voltage across the parallel section.**
- 0.2 × 15 = 3 V

**20d) Calculate the current through one 30 Ω resistor.**
- 3 ÷ 30 = 0.1 A